

Class: XI_STU_IC/IR_IIT_B1 & B2

JEE ADVANCED-2022-P1-Model

Date: 17-03-24

Time: 3hrs

PTA-02

Max. Marks: 180

IMPORTANT INSTRUCTIONS

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 –8)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 9 –14)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 15 – 18)	Questions with Matrix Matching	3	1	4	12
Total				18	60

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 –26)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 27 –32)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 33 – 36)	Questions with Matrix Matching	3	1	4	12
Total				18	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 –44)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 45 –50)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 51 – 54)	Questions with Matrix Matching	3	1	4	12
Total				18	60

Exam Syllabus

TOPIC

MATHS	Binomial Theorem, Permutations and combinations, Straight line and pair of straight lines, Parabola, Ellipse, Hyperbola
PHYSICS	Mechanical properties of solids, mechanical properties of Fluids, Thermal properties of matter, KTG, Thermodynamics, Oscillations, waves
CHEMISTRY	S-block elements, p-block elements: 13 th group elements, 14 th group elements Organic Chemistry: Some basic principles and techniques Basic concepts of organic chemistry, Hydrocarbons

SECTION – I
(NUMERICAL ANSWER TYPE)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/roundoff** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the correct numerical value is entered;
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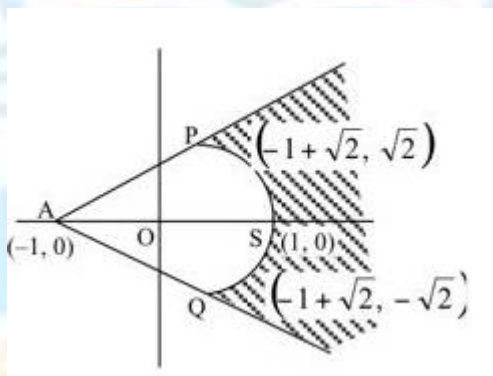
1. If the coefficients of x^7 and x^8 in the expansion of $\left(2 + \frac{x}{3}\right)^n$ are equal, then the value of n is
2. Given the coefficient of x^{21} in the expansion of $\left(x^2 + \frac{3a}{x}\right)^{15}$ is 3640, then the value of $\frac{3a}{8}$ is
3. If ${}^nP_r = {}^nP_{r+1}$ and ${}^nC_r = {}^nC_{r-1}$, then the value of $\frac{n}{r}$ is
4. The number of even divisors of the number $N = 12600$ is
5. The number of points on the ellipse $\frac{x^2}{50} + \frac{y^2}{20} = 1$ from which a pair of perpendicular tangents is drawn to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is
6. If $y = mx + c$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, having eccentricity 5, then the least positive integral value of m is
7. The area of the quadrilateral whose vertices are A(1, 1), B(3, 4), C(5, -2) and D(4, -7) is (in sq. units)
8. If the locus of mid points of portion of tangents intercepted between axes of hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is $\frac{A}{x^2} - \frac{9}{y^2} = B$, then $(A + B)$ is equal to

SECTION – II
(ONE OR MORE ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -2 In all other cases.

9. Parabola $y^2 + 3 = 2(2x + y)$ has
- (A) vertex at $\left(\frac{1}{2}, 1\right)$ and axis parallel to x -axis
- (B) vertex at $\left(1, \frac{1}{2}\right)$ and axis parallel to x -axis
- (C) vertex at $\left(\frac{1}{2}, 1\right)$ and focus at $\left(\frac{3}{2}, 1\right)$
- (D) vertex at $\left(\frac{1}{2}, 1\right)$ and axis parallel to y -axis
10. Consider the parabola $y^2 = 12x$. Then
- (A) line $y = 3x + 1$ is a tangent to the parabola
- (B) line $y = 2x - 36$ is a normal to the parabola
- (C) line $2x + y = 6$ is a chord of contact for some point on directrix of the parabola
- (D) line $x - 2y + 12 = 0$ subtends a right angle at the vertex of the parabola
11. PSQ is an arc of a circle as shown, then the shaded region will always satisfy



- (A) $z : |z+1| > 2; |\arg(z+1)| < \frac{\pi}{2}$
- (B) $z : |z+1| > 2; |\arg(z+1)| < \frac{\pi}{4}$
- (C) $z : |z-1| < 2; |\arg(z+1)| < \frac{\pi}{2}$
- (D) $z : |z-1| > 2; |\arg(z+1)| < \frac{\pi}{4}$
12. The equation of the straight line passing through (4, 3) and making intercepts on the co-ordinate axes whose sum is -1, is
- (A) $\frac{x}{-2} + \frac{y}{1} = -1$
- (B) $\frac{x}{-2} + \frac{y}{1} = 1$



(C) $\frac{x}{2} + \frac{y}{3} = 1$

(D) $\frac{x}{2} - \frac{y}{3} = 1$

13. The Complex number z satisfying the equations $|z + \bar{z}| + |z - \bar{z}| = 2$ and $|iz - 1| + |z - i| = 2$ simultaneously are

(A) i (B) $-i$ (C) $\frac{1}{i}$ (D) $\frac{1}{i^3}$

14. The minimum area of the triangle formed by the tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the coordinate axes is

(A) ab sq. units (B) $\frac{a^2 + b^2}{2}$ sq. units
 (C) $\frac{(a+b)^2}{2}$ sq. units (D) $\frac{a^2 + ab + b^2}{3}$ sq. units

SECTION – III (MATRIX MATCHING ANSWER TYPE)

- This section contains FOUR (04) Matching List Sets.
- Each set has ONE Multiple Choice Question.
- Each set has TWO lists: List-I and List-II.
- List-I has Four entries (I), (II), (III) and (IV) and List-II has Five entries (P), (Q), (R), (S), (T) and (U).
- FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks : +3 ONLY if the option corresponding to the correct combination is chosen;
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks : -1 In all other cases.

15. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .

- (i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.
 (ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.
 (iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.
 (iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.

List-I		List-II	
I)	The value of α_1 is	P)	136
II)	The value of α_2 is	Q)	189

III)	The value of α_3 is	R)	192
IV)	The value of α_4 is	S)	200
		T)	381
		U)	461

The correct option is

- (A) $I \rightarrow S; II \rightarrow U; III \rightarrow Q; IV \rightarrow P$
 (B) $I \rightarrow P; II \rightarrow S; III \rightarrow Q; IV \rightarrow R$
 (C) $I \rightarrow S; II \rightarrow U; III \rightarrow T; IV \rightarrow Q$
 (D) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow P$

16. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, be a hyperbola in the xy -plane whose conjugate axis LM subtends an angle of 60° at one of its vertices N . Let the area of the triangle LMN be $4\sqrt{3}$.

List-I		List-II	
I)	The length of the conjugate axis of H is	P)	8
II)	The eccentricity of H is	Q)	$\frac{4}{\sqrt{3}}$
III)	The distance between the foci of H is	R)	$\frac{2}{\sqrt{3}}$
IV)	The length of the latus rectum of H is	S)	4

The correct option is

- (A) $I \rightarrow S; II \rightarrow Q; III \rightarrow P; IV \rightarrow R$ (B) $I \rightarrow S; II \rightarrow R; III \rightarrow P; IV \rightarrow Q$
 (C) $I \rightarrow S; II \rightarrow P; III \rightarrow R; IV \rightarrow Q$ (D) $I \rightarrow R; II \rightarrow S; III \rightarrow Q; IV \rightarrow P$

17. Match the following.

List-I		List-II	
I)	Locus of point of intersection of $x = at^2, y = 2at$	P)	$x^2 + y^2 = 2a^2$
II)	Director circle of $x^2 + y^2 = a^2$	Q)	$y^2 = 4ax$
III)	Locus of point of intersection of the lines $x \cos \theta = y \cot \theta = a$	R)	$x^2 + y^2 = ax$
IV)	The locus of mid point of the chords of the circle $x^2 + y^2 - 2ax = 0$ passing through the origin	S)	$x^2 - y^2 = a^2$

- (A) $I \rightarrow S; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$ (B) $I \rightarrow P; II \rightarrow S; III \rightarrow Q; IV \rightarrow R$
 (C) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow R$ (D) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow P$

18. Match the following lists :

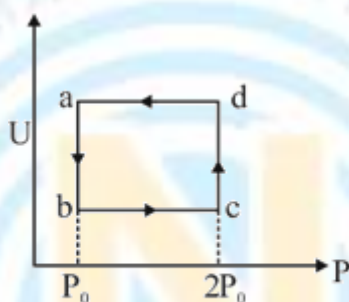
List-I		List-II	
I)	If $ z_1 + z_2 = z_1 - z_2 $, then	P)	$I_m = (z_1 \bar{z}_2) = 1$
II)	If $w_1 = a + ib; w_2 = p + iq, w_1 = w_2 = 1$ such that $I_m = (w_1 \bar{w}_2) = 1$. Now if $z_1 = a + ip$ and $z = b + iq$ then	Q)	$\frac{z_1}{z_2}$ is purely real.
III)	If $ z_1 + z_2 ^2 = z_1 ^2 + z_2 ^2$, then	R)	$\frac{iz_1}{z_2}$ is purely real.
IV)	If $ z_1 + z_2 = z_1 + z_2 $, then	S)	$\left \arg \left(\frac{z_1}{z_2} \right) \right = \frac{\pi}{2}$

- (A) $I \rightarrow R, S; II \rightarrow P; III \rightarrow R, S; IV \rightarrow Q$ (B) $I \rightarrow S; II \rightarrow P; III \rightarrow R, S; IV \rightarrow Q$
 (C) $I \rightarrow R, S; II \rightarrow Q; III \rightarrow S; IV \rightarrow Q$ (D) $I \rightarrow S; II \rightarrow P; III \rightarrow R, IV \rightarrow Q$

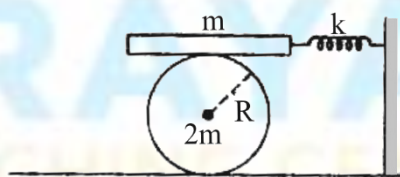
SECTION – I
(NUMERICAL ANSWER TYPE)

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19. Figure shows the variation of internal energy (U) with the pressure (P) of 2.0 moles of gas in cyclic process $abcd$. The temperatures of gas at c and d are 300 K and 500 K, respectively. The heat absorbed by the gas during the process is $\alpha(100R\ln 2)$. Find the value of α .



20. A uniform plank of mass $m = 1$ kg, free to move in the horizontal direction only, is placed at the top of a solid cylinder of mass $2m$ and radius R . The plank is attached to a fixed wall by means of a light spring of spring constant $k = 7$ N/m. There is no slipping between the cylinder and the plank, and between the cylinder and the ground. The angular frequency of small oscillations of the system is



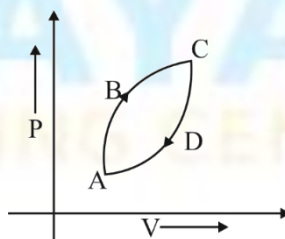
21. Two wires of same material and length but diameter in the ratio 1 : 2 are stretched by the same force. The potential energy per unit volume for the two wires when stretched will be in the ratio x : y . Find $x + y$.
22. If a capillary tube of radius r is immerse in water, then mass of water risen in capillary is M . If the radius of capillary be doubles, the mass of water risen in the capillary will be λM , find λ .
23. If the terminal speed of a sphere of gold (density = 19.5 kg/m^3) is 0.2 m/s in a viscous liquid (density = 1.5 kg/m^3), find the terminal speed of a sphere of silver (density = 10.5 kg/m^3) of the same size in the same liquid

24. The ratio of densities of two substances is 2 : 3 and that of specific heats is 1 : 2. The ratio of thermal capacities per unit volume is 1 : k, find k.
25. The percentage efficiency of a Carnot engine working between 800 K and 500 K is
26. A freely falling body takes 2 seconds to reach the ground on a planet when it is dropped from a height of 8 m. If the period of a simple pendulum is 2 seconds on that planet then its length is (in cm) (Use $\pi^2 = 10$).

SECTION – II (ONE OR MORE ANSWER TYPE)

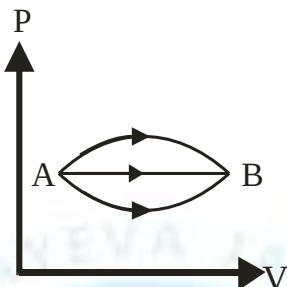
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27. The x-coordinate of a particle moving on x-axis is given by $x = 3\sin 100t + 8\cos^2 50t$ where x is in cm and t is time in seconds. Which of the following is/are correct about this motion.
- (A) The motion of the particle is not S.H.M
- (B) The amplitude of the S.H.M of the particle is 5 cm
- (C) The amplitude of the resultant S.H.M is $\sqrt{73}$ cm
- (D) The maximum displacement of the particle from the origin is 9 cm
28. Figure shows the P-V diagram of a cyclic process. If dQ is the heat energy supplied to the system, dU is change in the internal energy of the system and dW is the work done by the system, then which of the following relations is/are correct



- (A) $dQ = dU - dW$ (B) $dU = 0$
- (C) $dQ = dW$ (D) $dQ = -dW$
29. In Newton's law of cooling $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, the constant k is proportional to:
- (A) surface area of the body (B) the specific heat of the body
- (C) the mass of the body (D) the emissivity of the body
30. In a plane progressive harmonic wave

- (A) phase difference between displacement and acceleration of particle is zero
 (B) phase difference between displacement and acceleration of particle is π
 (C) phase difference between displacement and velocity of particle is $\pi/2$
 (D) phase difference between velocity and acceleration of particle is $\pi/2$
31. A gas undergoes change in its state from position A to position B, via three different paths as shown in the fig. Select the correct alternatives



- (A) Change in internal energy in all the three paths is equal
 (B) In all the three paths, heat is absorbed by the gas
 (C) Heat absorbed/released by the gas is maximum in path (1)
 (D) Temperature of the gas first increases and then decreases continuously in path (2)
32. When a wire is stretched to double its length
- (A) Its radius is halved
 (B) Strain is unity
 (C) Stress is equal to young's modulus
 (D) Young's modulus is equal to twice the elastic energy per unit volume

SECTION – III (MATRIX MATCHING ANSWER TYPE)

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33. A capillary tube of glass of radius R is placed inside a tub containing water. Capillary rise is h . Assume that $2h$ length is outside the water. Column I contain conditions and Column – II contains the corresponding capillary rise. Contact angle for glass and water is 0° .

Column – I	Column – II
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I)	If the arrangement is placed in an elevator going up with acceleration $g/2$	P)	$3h/2$
II)	If the whole arrangement is falling freely	Q)	$2h$
III)	If a glass rod of radius $R/2$ is inserted symmetrically in the capillary tube	R)	$2h/3$
IV)	If water in tub is replaced by fluid having density double and surface tension 3 times the values for water keeping other things same.	S)	Water spills out of the capillary

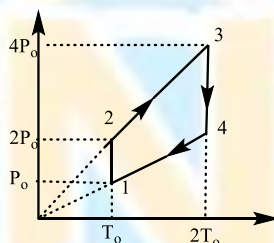
(A) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow P$

(B) $I \rightarrow R; II \rightarrow S; III \rightarrow Q; IV \rightarrow P$

(C) $I \rightarrow R; II \rightarrow Q; III \rightarrow P; IV \rightarrow P$

(D) $I \rightarrow P; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$

34. Read the following and write the correct pairs. Pressure – temperature graph of a thermodynamic process is shown (ρ -density).



Column I		Column II	
(I)	ρ -V graph	(P)	
(II)	P-V graph	(Q)	
(III)	ρ -P graph	(R)	
(IV)	T-V graph	(S)	

(A) $I \rightarrow P$; $II \rightarrow Q$; $III \rightarrow R$; $IV \rightarrow S$

(B) $I \rightarrow R$; $II \rightarrow P$; $III \rightarrow Q$; $IV \rightarrow S$

(C) $I \rightarrow R$; $II \rightarrow Q$; $III \rightarrow P$; $IV \rightarrow S$

(D) $I \rightarrow S$; $II \rightarrow P$; $III \rightarrow Q$; $IV \rightarrow S$

35. In the following question **column – I** represent some physical quantities & **column – II** represents their units, match them

Column – I		Column – II	
I)	Coefficient of linear expansion	P)	Cal/°C
II)	Water equivalent	Q)	gm
III)	Heat capacity	R)	(°C) ⁻¹
IV)	Specific heat	S)	Cal/g°C

(A) $I \rightarrow R$; $II \rightarrow Q$; $III \rightarrow P$; $IV \rightarrow S$

(B) $I \rightarrow R$; $II \rightarrow Q, S$; $III \rightarrow Q$; $IV \rightarrow P$

(C) $I \rightarrow R$; $II \rightarrow Q$; $III \rightarrow P$; $IV \rightarrow P$

(D) $I \rightarrow P$; $II \rightarrow R$; $III \rightarrow Q$; $IV \rightarrow P$

36. In the column – I, a system is described in each option and corresponding time period is given in the column – II. Suitably match them

Column – I		Column – II	
I)	A simple pendulum of length ' ℓ ' oscillating with small amplitude in a lift moving down with retardation $g/2$	P)	$T = 2\pi\sqrt{\frac{2\ell}{3g}}$
II)	A block attached to an end of a vertical spring, whose other end is fixed to the ceiling of a stationary lift, stretches the spring by length ' ℓ ' in equilibrium. It's time period when lift moves up with an acceleration $g/2$ is	Q)	$T = 2\pi\sqrt{\frac{\ell}{g}}$
III)	The time period of small oscillation of a uniform rod of length ' ℓ ' smoothly hinged at one end. The rod oscillates in vertical plane.	R)	$T = 2\pi\sqrt{\frac{2\ell}{g}}$
IV)	A cubical block of edge ' ℓ ' and specific gravity $\frac{1}{2}$ is in equilibrium with some volume inside water filled in a large fixed container. Neglect viscous forces and surface tension. The time period of small oscillations of the block in vertical direction is	S)	$T = 2\pi\sqrt{\frac{\ell}{2g}}$



- (A) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow P$
(B) $I \rightarrow R; II \rightarrow Q, S; III \rightarrow Q; IV \rightarrow P$
(C) $I \rightarrow P; II \rightarrow Q; III \rightarrow P; IV \rightarrow S$
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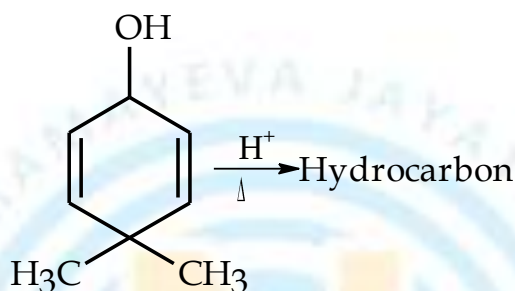


NARAYANA
COACHING CENTER

SECTION – I
(NUMERICAL ANSWER TYPE)

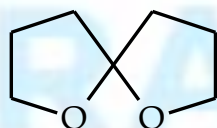
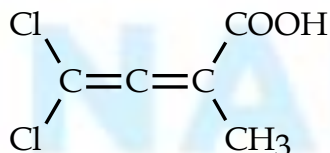
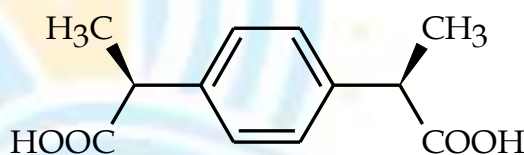
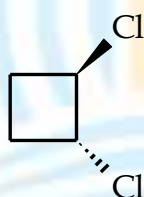
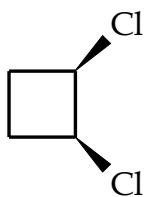
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37. Number of mono chloro derivatives of Aromatic Hydrocarbon is....

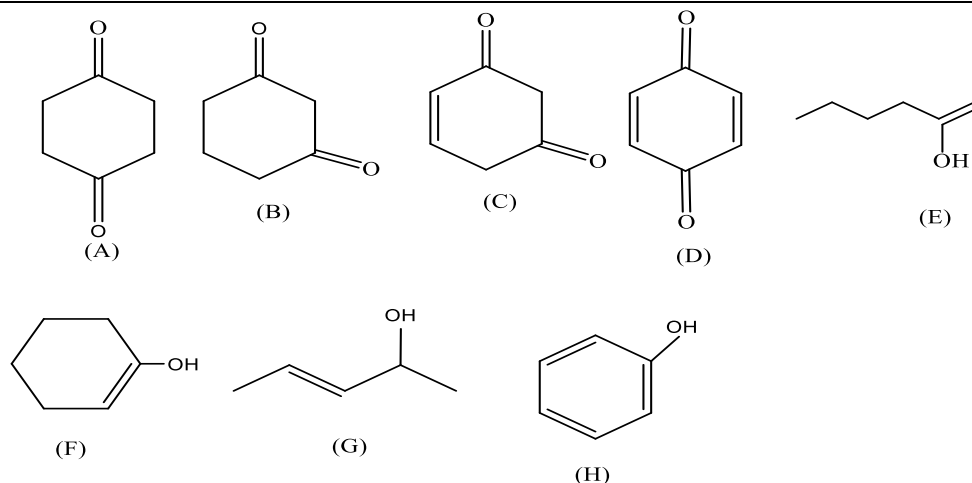


38. How many esters (excluding stereo isomers) are possible with Molecular formula, $C_5H_{10}O_2$.

39. Number of chiral compounds is



40. The following compound $(CH_3)_x Si(Cl)_y$ on hydrolysis followed by condensation gives a branched chain silicone. What is the value of y ?
41. How many of the following contains 3c-2e (three centred – two electron) bonds?
 $BeCl_2$ (solid), BeH_2 (solid), Al_2Cl_6 , B_2H_6 , $BeCl_2$ (vapour)
42. How many of the following can exhibit keto-enol tautomerism.



43. Choose the number of correct statement (s)
- (I) N-ethyl-N-methyl-1-propanamine is optically active
 (II) Cresol is more acidic than anisole
 (III) Pyrrole is more basic than pyridine
 (IV) butanone & butanol are constitutional isomers
44. How many of the following statements are correct regarding borax?
- (1) It consist of two six membered rings fused with each other
 (2) It consist of 3 and 4 coordinated Boron atoms
 (3) Its aqueous solution is a buffer with $\text{pH} > 7$
 (4) It is a primary standard for titrating acids
 (5) Its neutralization point with HCl can be determined by methyl orange

SECTION – II

(ONE OR MORE ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;
 Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
 Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
 Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks : -2 In all other cases.

45. I) $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH} = \text{CH}_2$
 II) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH} - \text{CH}_3$
 III) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH}_3$

The number of possible geometric isomers for the above compounds respectively are

- (A) 0,2,4 (B) 2,2,4 (C) 0,3,3 (D) 0,2,3

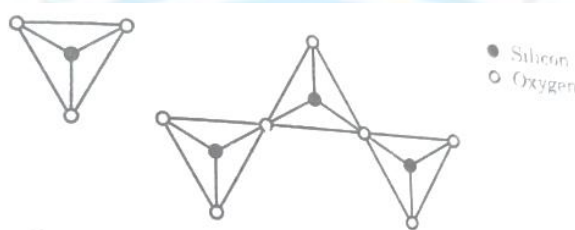
46. The statements about the given compound $\text{HO}-\text{C}_6\text{H}_4-\text{CH}=\text{CH}-\underset{\text{OH}}{\text{CH}}-\text{COOH}$ is/are

- (A) The compound can show geometrical isomerism
- (B) The compound can show optical isomerism
- (C) The compound can have position isomerism
- (D) Correct IUPAC name of this compound is 2-hydroxy-4-(4-hydroxyphenyl) but-3-enoic acid

47. The alkali metals dissolve in NH_3 it is found that

- (A) The dilute solution is blue colour but colour changes to bronze with increasing concentration
- (B) The blue colour is due to the presence of both solvated electrons and solvated ammonium ions.
- (C) The blue solution is paramagnetic
- (D) The pure metal is easily recovered when the ammonia evaporates.

48. The following pictures represent various silicate anions. Their formulae are respectively:

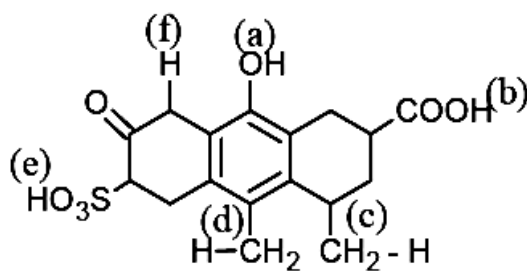


- (A) SiO_3^{2-} $\text{Si}_3\text{O}_7^{2-}$ (B) SiO_4^{4-} $\text{Si}_3\text{O}_{10}^{8-}$ (C) SiO_4^{2-} $\text{Si}_3\text{O}_9^{2-}$ (D) SiO_3^{4-} $\text{Si}_3\text{O}_7^{8-}$

49. Select the correct statement(s):

- (A) Carbon has pronounced ability to form $p\pi-p\pi$ multiple bonds to itself and to other elements like O and N
- (B) In diamond each carbon atom is linked tetrahedrally to four other carbon atoms by sp^3 hybridization
- (C) Graphite has planar hexagonal layers of carbon atoms held together by weak Van der Waal's forces
- (D) Out of CO_2 , SnO_2 , PbO_2 ; SnO_2 is amphoteric and PbO_2 is an oxidizing agent.

50. Arrange the following in increasing order of acidic strength.



(A) $c < f < d < a < b < e$

(B) $f < d < c < b < e < a$

(C) $f < c < d < b < e < a$

(D) $c < d < f < a < b < e$

SECTION – III (MATRIX MATCHING ANSWER TYPE)

- This section contains FOUR (04) Matching List Sets.
- Each set has ONE Multiple Choice Question.
- Each set has TWO lists: List-I and List-II.
- List-I has Four entries (I), (II), (III) and (IV) and List-II has Five entries (P), (Q), (R), (S) and (T).
- FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 ONLY if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

51.

Column-I		Column-II	
I)		P)	No of resonance structures are even numbered(kekule only)
II)		Q)	No of resonance structures are odd numbered(kekule only)
III)		R)	All C-C bond lengths are not equal
IV)		S)	Mono chloroderivatives are even numbered

(A) I → Q; II → Q,S; III → P,S; IV → Q,S

(B) I → P; II → Q,R,S; III → P,R; IV → Q,R

(C) $I \rightarrow Q$; $II \rightarrow Q, S$; $III \rightarrow P, S$; $IV \rightarrow R$

(D) $I \rightarrow P$; $II \rightarrow Q, S$; $III \rightarrow Q, S$; $IV \rightarrow Q, R$

52.

Column-I		Column-II	
I)	$\text{Borax} \xrightarrow{\Delta}$	P)	SiO_2
II)	$\text{B}_2\text{H}_6 + \text{H}_2\text{O} \longrightarrow$	Q)	H_2SiO_3
III)	$\text{B}_2\text{H}_6 + \text{Excess of } \xrightarrow{\Delta}$	R)	BN
IV)	$\text{SiCl}_4 \xrightarrow[(ii) 1000^\circ\text{C}]{(i) \text{H}_2\text{O}}$	S)	$\text{NaBO}_2 + \text{B}_2\text{O}_3$
		T)	H_3BO_3

(A) $I \rightarrow Q, R, S$; $II \rightarrow Q, S$; $III \rightarrow P, R$; $IV \rightarrow P, R, S$

(B) $I \rightarrow P, R, S$; $II \rightarrow Q, S$; $III \rightarrow P, R$; $IV \rightarrow P, S$

(C) $I \rightarrow P, R, S$; $II \rightarrow Q, S$; $III \rightarrow P, R$; $IV \rightarrow Q, R, S$

(D) $I \rightarrow P, S$; $II \rightarrow Q, S$; $III \rightarrow P, R, S$; $IV \rightarrow P, R, S$

53.

Column-I		Column-II	
(I)	$\text{Borax} \xrightarrow{\Delta}$	(P)	SiO_2
(II)	$\text{B}_2\text{H}_6 + \text{H}_2\text{O} \longrightarrow$	(Q)	H_2SiO_3
(III)	$\text{B}_2\text{H}_6 + \text{Excess of } \text{NH}_3 \xrightarrow{\Delta}$	(R)	BN
(IV)	$\text{SiCl}_4 \xrightarrow[(ii) 1000^\circ\text{C}]{(i) \text{H}_2\text{O}}$	(S)	$\text{NaBO}_2 + \text{B}_2\text{O}_3$
		(T)	H_3BO_3

(A) $I \rightarrow P$; $II \rightarrow R$; $III \rightarrow S$; $IV \rightarrow T$

(B) $I \rightarrow Q$; $II \rightarrow R$; $III \rightarrow S$; $IV \rightarrow T$

(C) $I \rightarrow S$; $II \rightarrow T$; $III \rightarrow R$; $IV \rightarrow Q$

(D) $I \rightarrow S$; $II \rightarrow T$; $III \rightarrow R$; $IV \rightarrow P$

54.

Column-I		Column-II	
(I)	CaH_2	(P)	Paramagnetic anion
(II)	K_2O_2	(Q)	Homodiatom, diamagnetic anion
(III)	KO_2	(R)	Neutral aqueous solution
(IV)	NaCl	(S)	Gives hydrogen on hydrolysis

(A) $I \rightarrow P$; $II \rightarrow R$; $III \rightarrow S$; $IV \rightarrow Q$

(B) $I \rightarrow Q$; $II \rightarrow R$; $III \rightarrow S$; $IV \rightarrow P$

(C) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ Q

 (D) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R

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